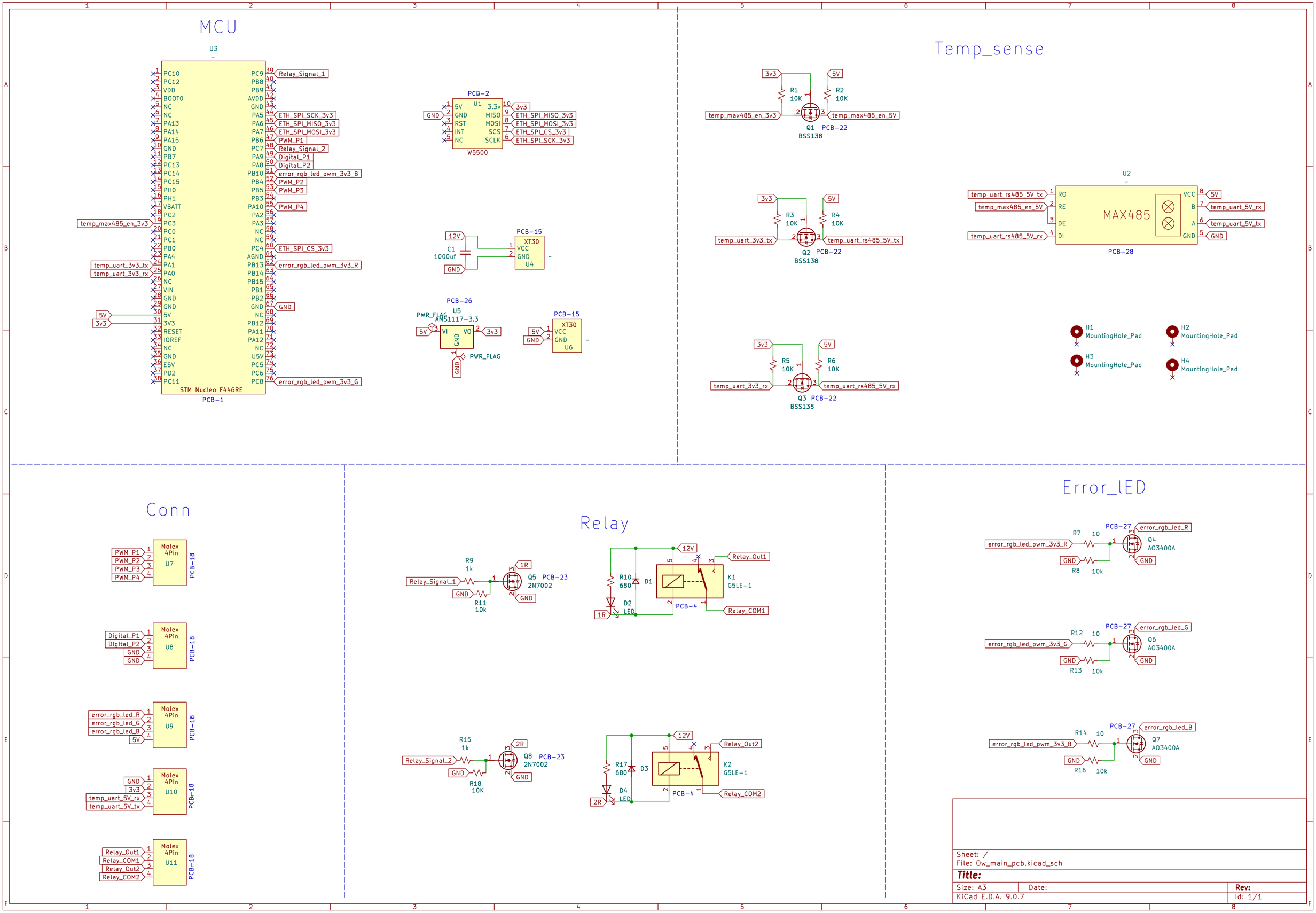
MCU
U3
STM Nucleo F446RE
PCB-1
PC10 PC12 VDD BOOT0 NC NC PA13 PA14 PA15 GND PB7 PC13 PC14 PC15 PH0 PH1 VBATT PC2 PC3 PC0 PC1 PB0 PA4 PA1 PA0 NC VIN GND GND 5V 3V3 RESET IOREF NC GND E5V PD2 PC11
PC9 PB8 PB9 AVDD GND PA5 PA6 PA7 PB6 PC7 PA9 PA8 PB10 PB4 PB5 PB3 PA10 PA2 PA3 NC NC PC4 AGND PB13 PB14 PB15 PB1 PB2 GND NC PB12 PA11 PA12 NC U5V PC5 PC6 PC8
temp_max485_en_3v3
temp_uart_3v3_tx
temp_uart_3v3_rx
5V
3v3
Relay_Signal_1
ETH_SPI_SCK_3v3
ETH_SPI_MISO_3v3
ETH_SPI_MOSI_3v3
PWM_P1
Relay_Signal_2
Digital_P1
Digital_P2
error_rgb_led_pwm_3v3_B
PWM_P2
PWM_P3
PWM_P4
ETH_SPI_CS_3v3
error_rgb_led_pwm_3v3_R
GND
error_rgb_led_pwm_3v3_G
PCB-2
U1
W5500
5V GND RST INT NC
3.3v MISO MOSI SCS SCLK
GND
3v3
ETH_SPI_MISO_3v3
ETH_SPI_MOSI_3v3
ETH_SPI_CS_3v3
ETH_SPI_SCK_3v3
PCB-15
XT30
VCC GND
U4
12V
C1
1000uf
GND
PCB-26
U5
AMS1117-3.3
PWR_FLAG
VI VO GND
5V
3v3
PCB-15
XT30
VCC GND
U6
Temp_sense
R1 10K
R2 10K
Q1 PCB-22
BSS138
temp_max485_en_3v3
temp_max485_en_5V
R3 10K
R4 10K
Q2 PCB-22
BSS138
temp_uart_3v3_tx
temp_uart_rs485_5V_tx
R5 10K
R6 10K
Q3 PCB-22
BSS138
temp_uart_3v3_rx
temp_uart_rs485_5V_rx
U2
MAX485
RO RE DE DI
VCC B A GND
temp_uart_rs485_5V_tx
temp_max485_en_5V
temp_uart_rs485_5V_rx
5V
temp_uart_5V_rx
temp_uart_5V_tx
GND
PCB-28
H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
H4 MountingHole_Pad
Conn
Molex 4Pin U7 PCB-18
PWM_P1 PWM_P2 PWM_P3 PWM_P4
Molex 4Pin U8 PCB-18
Digital_P1 Digital_P2 GND GND
Molex 4Pin U9 PCB-18
error_rgb_led_R error_rgb_led_G error_rgb_led_B 5V
Molex 4Pin U10 PCB-18
GND 3v3 temp_uart_5V_rx temp_uart_5V_tx
Molex 4Pin U11 PCB-18
Relay_Out1 Relay_COM1 Relay_Out2 Relay_COM2
Relay
R9 1k
Relay_Signal_1
1R
Q5 PCB-23
2N7002
GND
R11 10k
R10 680
D1
D2 LED
12V
Relay_Out1
K1
G5LE-1
PCB-4
Relay_COM1
R15 1k
Relay_Signal_2
2R
Q8 PCB-23
2N7002
R18 10K
R17 680
D3
D4 LED
Relay_Out2
K2
G5LE-1
Relay_COM2
Error_LED
PCB-27
R7 10
error_rgb_led_pwm_3v3_R
error_rgb_led_R
Q4
AO3400A
R8 10k
R12 10
error_rgb_led_pwm_3v3_G
error_rgb_led_G
Q6
AO3400A
R13 10k
R14 10
error_rgb_led_pwm_3v3_B
error_rgb_led_B
Q7
AO3400A
R16 10k
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Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 1/1